

BOB WEEK

Curriculum Vitae

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EDUCATION

- 2020 **PhD Bioinformatics & Computational Biology** S.L. Nuismer Lab, IBEST, University of Idaho
Dissertation focused on modeling eco-evolutionary processes and developing statistical methods
- 2015 **BS Mathematics** University of Idaho
Traditional math degree with electives in electrical engineering

PEER-REVIEWED PUBLICATIONS

- 2026 **Stochastic Eco-Evolutionary Dynamics of Multivariate Traits** *Journal of Theoretical Biology*
Week, B. doi:10.1016/j.jtbi.2026.112428
- 2025 **Quantitative Genetics of Microbiome Mediated Traits** *Evolution*
Week, B.; Ralph, P.L.; Tavalire, H.F.; Cresko, W.A.; Bohannan, B.J.M. doi:10.1093/evolut/qpaf171
- 2025 **Applying Evolutionary Theory to Understand Host-Microbiome Evolution** *Nature EcoEvo*
Week, B.; Russel, S.L.; Schulenburg, H.; Bohannan, B.J.M.; Bruijning, M. doi:10.1038/s41559-025-02846-w
- 2023 **Host-Parasite Coevolution in Continuous Space Leads to Variation in Local Adaptation Across Spatial Scales** *The American Naturalist*
Week, B.; Bradburd, G.S. doi:10.1086/727470
- 2022 **Uncovering Cryptic Coevolution** *The American Naturalist*
Nuismer, S.L.; Week, B.; Harmon, L.J. doi:10.1086/717436
- 2021 **A White Noise Approach to Evolutionary Ecology** *Journal of Theoretical Biology*
Week, B.; Nuismer, S.L.; Harmon, L.J.; Krone, S.M. doi:10.1016/j.jtbi.2021.110660
- 2021 **Coevolutionary Arms Races and the Conditions for the Maintenance of Mutualism** *The American Naturalist*
Week, B.; Nuismer, S.L. doi:10.1086/714274
- 2021 **A Unified Model of Species Abundance, Genetic Diversity, and Functional Diversity Reveals the Mechanisms Structuring Ecological Communities** *Molecular Ecology Resources*
Overcast, I.; Ruffley, M.; Rosindell, J.; Harmon, L.; Borges, P.; Emerson, B.; Etienne, R.S.; Gillespie, R.; Krehenwinkel, H.; Mahler, L.; Massol, F.; Parent, K.; Patiño, J.; Peter, B.; Week, B.; Wagner, C.; Hickerson, M.J.; Rominger, A. doi:10.1111/1755-0998.13514
- 2019 **Identifying Models of Trait-Mediated Community Assembly using Random Forests and Approximate Bayesian Computation** *Ecology and Evolution*
Ruffley, M.; Peterson, K.; Week, B.; Tank, D.; Harmon, L.J. doi:10.1002/ece3.5773
- 2019 **Approximate Bayesian Estimation of Coevolutionary Arms Races** *PLOS Computational Biology*
Nuismer, S.L.; Week, B. doi:10.1371/journal.pcbi.1006988
- 2019 **The Measurement of Coevolution in the Wild** *Ecology Letters*
Week, B.; Nuismer, S.L. doi:10.1111/ele.13231
- 2018 **Coevolution Slows the Disassembly of Mutualistic Communities** *The American Naturalist*
Nuismer, S.L.; Week, B.; Aizen, M. doi:10.1086/699218

PREPRINTS

- 2024 **The Evolution of Microbiome-Mediated Traits** *bioRxiv*
Week, B.; Morris, A.H.; Bohannan, B.J.M. doi:10.1101/2024.03.29.587374

MANUSCRIPTS IN PREPARATION

- 2026 **Microbially Assisted Rescue of Host Populations** *Manuscript in preparation*
Week, B.; Schulenburg, H.; Uecker, H. doi:

AWARDS

2024 – 2027	EU/DFG Postdoctoral Fellowship The "Kiel Training for Excellence" programme is cofunded by the Marie Skłodowska-Curie Actions from the European Commission's Horizon Europe programme (project number: 101081480) and by Kiel University	KiTE, Kiel University
2018 – 2019	Bioinformatics & Computational Biology Fellowship Project aimed to model the duration of coevolutionary associations	IBEST, University of Idaho
2017-2018	Bioinformatics & Computational Biology Fellowship Project aimed to develop a statistical method to measure coevolution in continuous space	IBEST, University of Idaho
2017	Paul Joyce Memorial BCB Fellowship Endowment Nominated by Professor Scott Nuismer because of my "love for mathematics and helping others to appreciate how it can be used to understand biological processes"	IBEST, University of Idaho
2013-2015	Undergraduate Research in Biology & Mathematics Efforts focused on developing a statistical method to measure coevolution in metapopulations	IBEST, University of Idaho

PROFESSIONAL EXPERIENCE

2024 – 2027	KiTE Postdoctoral Research Fellow Establishing theoretical foundations for the study of microbiome mediated trait dynamics	H. Schulenburg Lab, Kiel University
2022 – 2024	Postdoctoral Research Fellow Extended evolutionary theory for traits jointly determined by host genotype and host microbiome	B.J.M. Bohannan Lab, University of Oregon
2020 – 2022	Postdoctoral Researcher Developed mathematical and computational approaches to understand coevolution in continuous space	G.S. Bradburd Lab, Michigan State University
2018	Visiting Scientist Field ecology training on estimating floral abundance and phenology, recording plant-pollinator interactions and estimating percent cover	P.J. CaraDonna Lab, Rocky Mountain Biological Laboratory

TEACHING EXPERIENCE

2026	Guest Lecturer Introduced coevolution to a masters student class, emphasizing ecological aspects and touching on the role of geography. <i>Slides linked here</i>	Kiel University, BIOL227 - Ecology, Evolution, and Genetics
2017	Teaching Assistant Taught the lab portion of a 300-level ecology and population biology course	University of Idaho, Department of Biological Sciences
2012 – 2014	Mathematics Tutor Part-time work at tutoring center supporting students taking a wide-range of coursework	Clark Community College, Mathematics Department

PRESENTATIONS

2026	The Response of Genetic Correlations to Drift - Talk	Aquavit Seminar - MPI, Plön
2026	Microbially Assisted Rescue of Host Populations - Poster	ISEMPH 2026 - Kiel, Germany
2026	The Response of G-matrices to Random Genetic Drift - Talk	MMEE 2026 - Cork, Ireland
2026	Quantitative Genetics of Microbiome Mediated Traits - Talk	BIRS - Banff, Canada
2025	Quantitative Genetics of Microbiome Mediated Traits - Talk	ESEB 2025 - Barcelona, Spain
2025	Quantitative Genetics of Microbiome Mediated Traits - Seminar	Theoretical Biology Department - MPI, Plön
2025	When is Microbial Rescue More or Less Effective than Genetic Rescue? - Poster	Evolutionary Rescue Workshop - MPI, Plön
2025	Microbiome-Mediated Host Adaptation: A Niche Construction Approach - Poster	Concepts in Evolution Workshop - MPI, Plön
2024	Host-Parasite Coevolution & Microbiome-Mediated Adaptation - Seminar	TransEvo Core Seminar - CAU, Kiel
2023	The Evolution of Microbiome-Mediated Traits - Talk	Symbiosis Theory Workshop - Eugene, Oregon

2023	Modeling Adaptation of Microbiome-Mediated Traits - Talk	EvoWibo - Port Townsend, Washington
2022	Host-Parasite Coevolution in Continuous Space - Poster	PEQG2022 - Pacific Grove, California
2021	Coevolutionary Arms Races and The Conditions for The Maintenance of Mutualism - Talk	AmNat2021 - Virtual
2020	A Bayesian Methodology for Estimating the Distribution of Coevolution within Ecological Communities - Talk	AmNat2020 - Pacific Grove, California
2018	The Measurement of Coevolution in Nature - Poster	EvoWibo - Port Townsend, Washington
2017	The Measurement of Coevolution in Mutualisms - Talk	Evolution - Portland, Oregon

SERVICE & LEADERSHIP

2022	Code Contributor Developed a nucleotide-based model of coevolution for SLiM. See §19.7 <i>here</i> .	SLiM 4.0 doi:10.1086/723601
2018-2019	Graduate Student Representative Represented graduate students in the Bioinformatics & Computational Biology program at institutional meetings	IBEST, University of Idaho
—	Manuscript Reviewer The American Naturalist, Ecology, Evolution, PCI Evol Biol, Population Ecology, Proceedings of The Royal Society B, Theoretical Population Biology, Molecular Biology & Evolution	

SOCIETIES

2021-Present	The International Society of Nonbinary Scientists	isnbs.org
2020-Present	The American Society of Naturalists	amnat.org

INTERESTS

I am broadly interested in collaborating on any scientific topic where my skills are useful. I am particularly interested in developing and formalizing models to clarify conceptual issues in population biology and community ecology.

SKILLS

Software:	$\text{\LaTeX}$, Python, R, Linux, Julia, Mathematica, SLURM, SLiM, C/C++
Statistics:	Modeling, Analysis, Inference, Methods Development
Math:	Linear Algebra, Dynamical Systems, Functional Analysis, Stochastic Processes